MPJ LAB - 2

PROBLEM STATEMENT

Write a prog in Java showing hirarchical inheritance with base class as Emp and derived classes as FullTimeEmployee and InternEmployee with methods DisplaySalary in base class and CalculateSalary in derived classes. Calc sal method will calculated as per increment given to fulltime and intern Emp. fulltime emp - 50%. hike, Intern emp - 25% hike. Display sal before and afte hike

OBJECTIVE

To study

- Inheritance in Java
- why to use inheritance
- types inheritance

THEORY

Inheritance in Java

It is an OOC in which one class acquires the properties and method of another class

It helps in code reuse and creates a parent - child relationship between classes

Use of Inheritance

- Reuse existing code, reducing duplication
- Improve maintainability of programs
- Support method overriding

FOR EDUCATIONAL USE

- Represent real-world relationships
- Increase code readability and organization

→ Types
- Single inheritance
  one child class inherits from one parent class
- Multilevel inheritance
  class inherits from another class which is already derived from another class
- Hierarchical Inheritance
  multiple child class from single parent
- Multiple Inheritance
  a class inherits from multiple interfaces
- Hybrid Inheritance
  Combination of 2 or more types of inheritance

CONCLUSION
Thus we have successfully implemented usage of Inheritance in Java

FAQ

Q1) Is multiple inheritance supported in Java? How is it achieved?
→ Multiple inheritance is NOT supported in Java
It is to avoid ambiguity problem such as Diamond Problem
Java supports multiple inheritance using interfaces

How it is achieved
- A class can implement multiple inheritance using implements keyword
  Interfaces contain abstract methods so there is no ambiguity

Q2) IS-A relationship in Java
→ represents inheritance in Java
It means that subclass is type of its superclass

Eg
Dog is animal
Student is person

In Java it is achieved using extends keyword & is verified using instance of operator

Q3) Are constructor & instance initialization block inherited to subclass
→ No, it is not inherited by subclass
Constructors are used to initialize objects, not to defined behaviour
Instance initialization block are executed during object creation but belong only to class in which they are defined

A subclass constructor can call superclass constructor using super()


Sundaram FOR EDUCATIONAL USE